

Practical Task Sheet 2

Modelling and Creation of the Data Warehouse

The fictitious beverage market chain *DrinkMix* will continue to serve as an example scenario for the practical task.

On the basis of the management requirements, various perspectives for product sales can be identified. On the one hand, articles are grouped into product categories 1 (alcoholic, non-alcoholic). Furthermore, it should be possible to analyze product sales by day, week, month, quarter and year. In addition, each branch is assigned to a geographical region, according to which it should also be possible to aggregate:

- branch 1 : east
- branch 2 : west
- branch 3 : north
- branch 4 : north
- branch 5 : east
- branch 6 : west

For all customers there is reference data to uniquely identify them:

- first name and family name
- adress : street, number, ZIP code, city
- gender
- date of birth

Customers should be able to be classified by gender. Furthermore, it should be possible to aggregate by age group.

- child : 0 – 15 years
- youngster : 16 – 19 years
- young adult : 20 – 29 years
- adult : 30 – 49 years
- old adult : over 50 years

For each sales item, the quantity should be stored in liters.

Task

This subproject focuses on the modeling and physical implementation of the data warehouse. For this purpose, a multidimensional data schema is to be created using the mE/R model. Subsequently, the mE/R schema is to be converted into a relational schema. The Star schema will be used for this purpose.

This shall then be transformed into a physical Oracle DBS schema and created in the corresponding group area.

For the graphical presentation of the schemas, any tools can be used again, but the submissions should be in PDF format. For the physical schema, submit the respective SQL DDL statements as SQL file.

An overview of all things to do in this task:

- analysis of the scenario description
- modeling of the data warehouse as mE/R schema
- transformation of the mE/R schema into a relational Star schema
- Implementation of the Star schema in the assigned group area in the Oracle database